

TANIYA NAUTIYAL

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PROFILE

Dynamic Developer with 1 year of experience in data analysis and machine learning, dedicated to delivering impactful solutions through technical expertise in Python, SQL, and web development. Skilled in transforming complex datasets into actionable insights and creating engaging web applications that enhance user experience. Proven ability to collaborate effectively within teams, applying critical thinking and detailed analysis to achieve project success. Committed to continuous learning and adapting to emerging technologies, ensuring readiness to tackle evolving business challenges.

EDUCATION

SRM-IST Aug 2023 - Aug 2027 (Expected)

Bachelor's of Technology - Computer Science Engineering , Delhi-NCR
CGPA: 7.76

Courses: Operating Systems, Data Structures, Analysis Of Algorithms, Artificial Intelligence, Machine Learning, Python, Computer-Networking, Database Management System, Data Science, Natural Language Processing, Object Oriented Design and Programming, Computer Graphics and Animation

SKILLS SUMMARY

Languages: Python, C++, SQL, MySQL
Developer Tools & Frameworks: Flask, Git/GitHub, Unreal Engine
Data Analysis & Design: MS-Excel, MS-Powerpoint, Matplotlib, NetworkX

EXPERIENCE

Intern,KraftonIndia Private Limited June 2025 - July 2025

- Designed Mystic Labyrinth, an accessible adventure-puzzle concept focused on calm, non-competitive exploration for players of all abilities, including neurodivergent users.
- Created game mechanics and level concepts with no fail states, gentle guidance, and flexible pacing to support diverse cognitive and sensory needs.
- Defined multi-platform experience (mobile, tablet, PC) with multiple input modes to accommodate varied physical and sensory accessibility requirements.
- Aligned design with inclusive gaming best practices, emphasizing sensory-friendly environments and customizable interaction options for players with disabilities

Intern, Defence Research and Development Organisation January 2026 - March 2026

- Co-developed a production-ready medical web application for automated brain tumor classification, fine-tuning a ResNet-50 deep learning model on 50,000+ labeled MRI scans using PyTorch.
- Achieved a 95%+ weighted precision and a weighted F1-score of 0.951 (ROC-AUC: 0.989) in identifying four distinct clinical categories: glioma, meningioma, pituitary, and no tumor.
- Engineered a secure three-tier architecture combining a responsive browser UI, a Flask application layer, and a containerized PyTorch inference engine deployed as a Docker REST endpoint.
- Optimized production inference latency using Gunicorn to achieve benchmarks of 30–50 ms on GPU (140–200 ms on CPU) with an enterprise-grade throughput of 15–25 requests per second at P95.
- Implemented HIPAA-compliant security layers and robust MLOps practices, incorporating zero-persistent image storage, TLS 1.2+ encryption, and automated CI/CD pipelines via GitHub Actions.

PROJECTS

HealthNexus- AI | Python, Scikit-Learn, Pandas, NumPy, Machine Learning

- Developed an AI-powered healthcare application to facilitate early disease detection and health monitoring
- Processed and analyzed structured medical data to generate actionable insights.
- Integrated machine learning models for prediction and classification of health conditions.
- Enhanced decision support accuracy through data preprocessing and feature engineering.

TACK - A Finance Tracker | Flask, Python, MySQL, HTML5/CSS3, JavaScript

- Developed a secure finance tracking web app using Flask for managing income and expenses.
- Implemented user authentication and role-based access control.
- Developed transaction management features with CRUD operations.
- Provided analytical insights into spending behavior using categorized data.
- Designed a responsive UI to enhance usability across devices.

Railway Network Site | Python, NetworkX, Matplotlib, Graph Theory Algorithms

- Developed Minimum Spanning Tree (MST) system for optimizing railway networks using Kruskal's and Prim's algorithms to connect stations with minimal total track length.
- Implemented interactive Python application with NetworkX for graph construction and Matplotlib for visualizing networks and MST results across small to large scales.
- Designed user input validation for stations/tracks, computed MST edges/weights, and verified performance on networks upto 50+ nodes with consistent outputs.
- Demonstrated practical graph theory applications in transportation infrastructure, supporting cost reduction and scalable network planning.

Online Railway Reservation System | MySQL, Relational Database Management , Database Design

- Developed Online Railway Reservation System using MySQL DBMS, enabling seamless ticket booking, cancellations, inquiries, and train schedule management for users and admins.
- Designed comprehensive database schema with ER diagrams, normalized tables (User, Train, Ticket, Payment, Class), constraints, joins, views, triggers, and cursors for efficient data handling and passenger count automation.
- Implemented concurrency control via locking protocols and recovery mechanisms like Write-Ahead Logging (WAL) to ensure data consistency during multi-user transactions and system failures.
- Created user-friendly interfaces with real-time updates and admin controls for ticket approval, supporting scalable rail travel management with future mobile/payment gateway integration.

VOLUNTEER EXPERIENCE

PR and Marketing Team- Hackhound

- Conducted an offline technical hackathon event with over 250 participants from across India.

ADDITIONAL INFORMATION

- **Languages:** English, Hindi